

Enterprise Security Assessment (ESA)

Assessment Data Gathering Instructions



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Read Guidance

- Start by reviewing each slide carefully.
- Choose your preferred export method: manual or script-based.
Note: Microsoft Secure Score (Microsoft Defender XDR) and Purview Compliance Manager must always be exported manually.
- Follow the steps provided in the relevant export method slides.
- At the end, review the checklist slide to ensure all required steps have been completed.

Data Sources

You need to export four files from the following sources

Defender for Cloud – Secure Score CSV File

The recommendations in Microsoft Defender for Cloud helps improve cloud security posture by aggregating security findings into a single score. Recommendations are grouped into security controls, which are logical groups of related security recommendations reflecting vulnerable attack surfaces

Defender for Cloud – MCSB CSV File

The Microsoft Cloud Security Benchmark (MCSB) provides prescriptive best practices and recommendations to help improve the security of workloads, data, and services on Azure and multi-cloud environments. This benchmark focuses on cloud-centric control areas with input from a set of holistic Microsoft and industry security guidance.

Microsoft Defender XDR CSV File

Microsoft Secure Score provides a holistic measure of an organization's security posture across various Microsoft services, including Microsoft 365 and Azure, focusing on identity, data, devices, apps, and infrastructure (for example, Entra ID, Teams, Exchange Online, SharePoint Online, and more).

Microsoft Purview Excel File

Microsoft Purview Compliance Manager improvement actions help organizations assess and manage their compliance posture across multicloud environments by providing recommended guidance aligned with data protection regulations and standards, with automated testing and continuous monitoring.

Minimum **Azure** Permissions:
Security Reader or
Reader

Minimum **Entra ID**
permission:
Global Reader

Minimum **Purview** permission:
Compliance Manager Reader
or **Entra ID Global Reader**

Data Export

Two Export Methods

Manual Export

- Suitable for a few subscriptions and/or a low number of recommendation rows.
- You may encounter download errors from the Azure Portal.

OR

Export via PowerShell Script

- Suitable for large number of subscriptions and/or big data sets.
- Easy-to-run repeatable process to refresh the Power BI report on a regular basis.
- Less error-prone.
- **Recommended method.**
- Script Location:
<https://github.com/microsoft/esa>

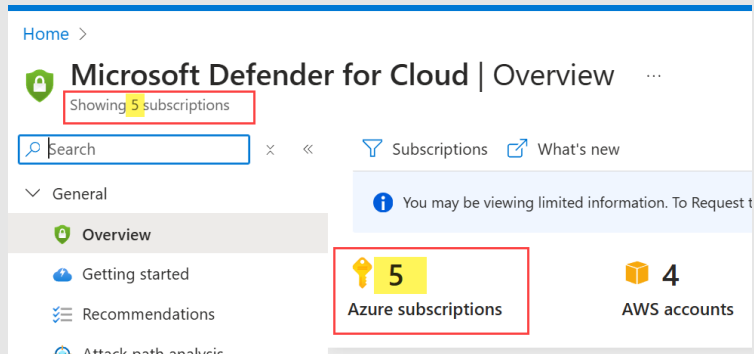
Note: Microsoft Defender XDR and Microsoft Purview recommendations need to be exported manually.

Manual Export

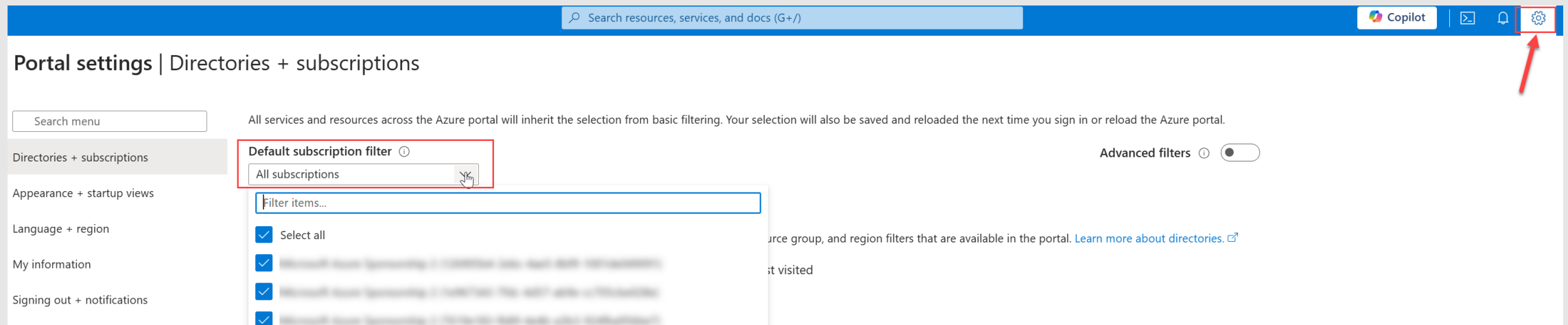
Instructions for Exporting Defender for Cloud
Recommendations

Review and filter selected subscriptions

In the Azure Portal, navigate to Defender for Cloud and review the selected subscriptions:



You may want to use the Azure Portal subscription filter to include or exclude subscriptions from the export, based on the scope of your assessment:



Export Defender for Cloud: Secure Score

In Defender for Cloud, select "Recommendations":

Home > Microsoft Defender for Cloud

Microsoft Defender for Cloud | Recommendations

Showing subscription [redacted]

Search [] × << Refresh Download CSV report Open query Governance report Guides & Feedback **Switch to classic view**

General

- Overview
- Setup
- Recommendations**

Scope: Azure subscriptions 1 AWS accounts 1 GCP projects 0 GitHub connectors 1 Azure containers 1

Defender CSPM

Recommendations by risk Risk based recommendations

Important:

Make sure to switch to Classic view before exporting recommendations, as only the classic view provides the correct format required for the assessment.

Export Defender for Cloud: Secure Score

Download **Secure Score recommendations** by selecting **Download CSV report**

The screenshot shows the Microsoft Defender for Cloud Recommendations page. At the top, there is a breadcrumb trail: Home > Microsoft Defender for Cloud | Recommendations >. Below this is the 'Recommendations' header. A red box highlights the 'Download CSV report' button, which is located next to 'Refresh', 'Open query', 'Governance report', and 'Guides'. Below the header, there is a message: 'You may be viewing limited information. To Request tenant-wide visibility, click here →'. The main section is titled 'Secure score recommendations' and 'All recommendations'. A red box highlights the 'Secure score' section, which displays a shield icon, '44%', and 'Secure score ⓘ'. To the right of this, it shows '153/247' and 'Active secure score recommendations'. Below the score, there is a search bar and filters for 'Recommendation status == None' and 'Severity =='. A list of recommendations is shown below, including 'Enable MFA', 'Secure management ports', and 'Remediate vulnerabilities'.

Important:

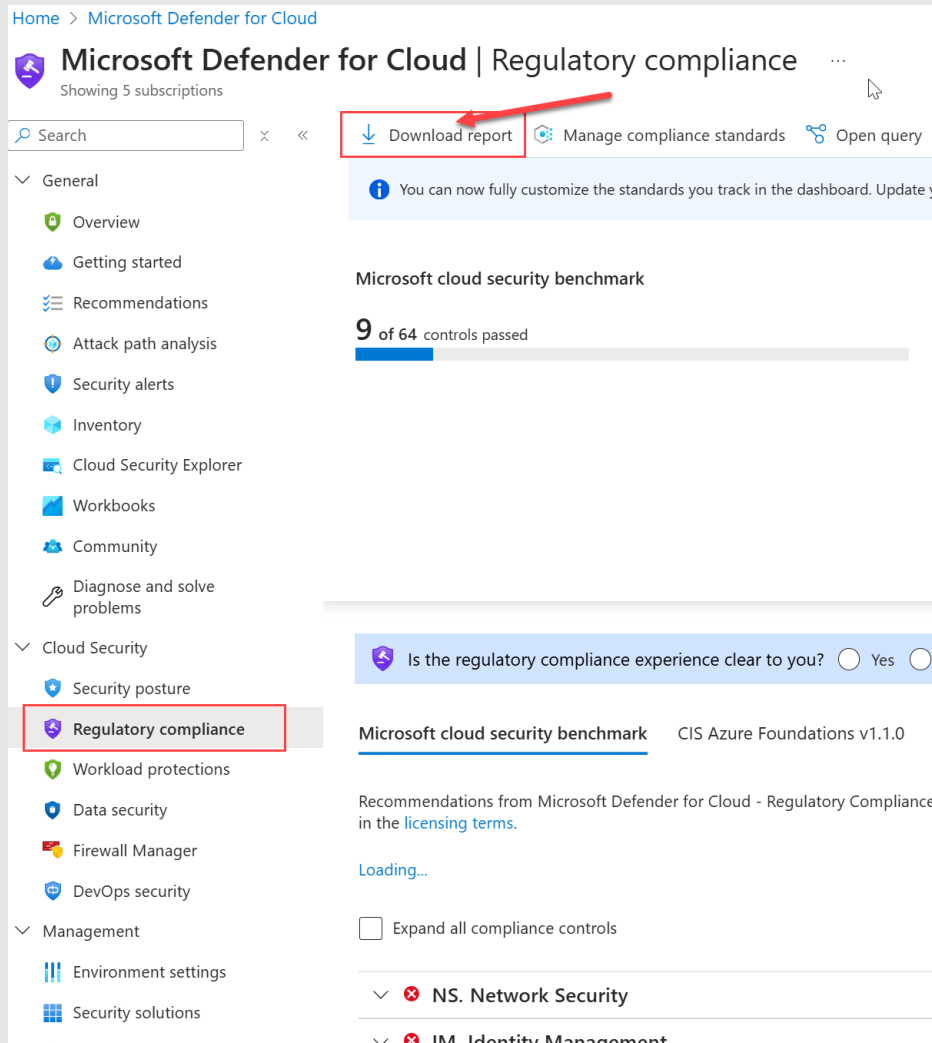
Make a note of the Secure Score value, as this number will also be shared with Microsoft.

Manual Export

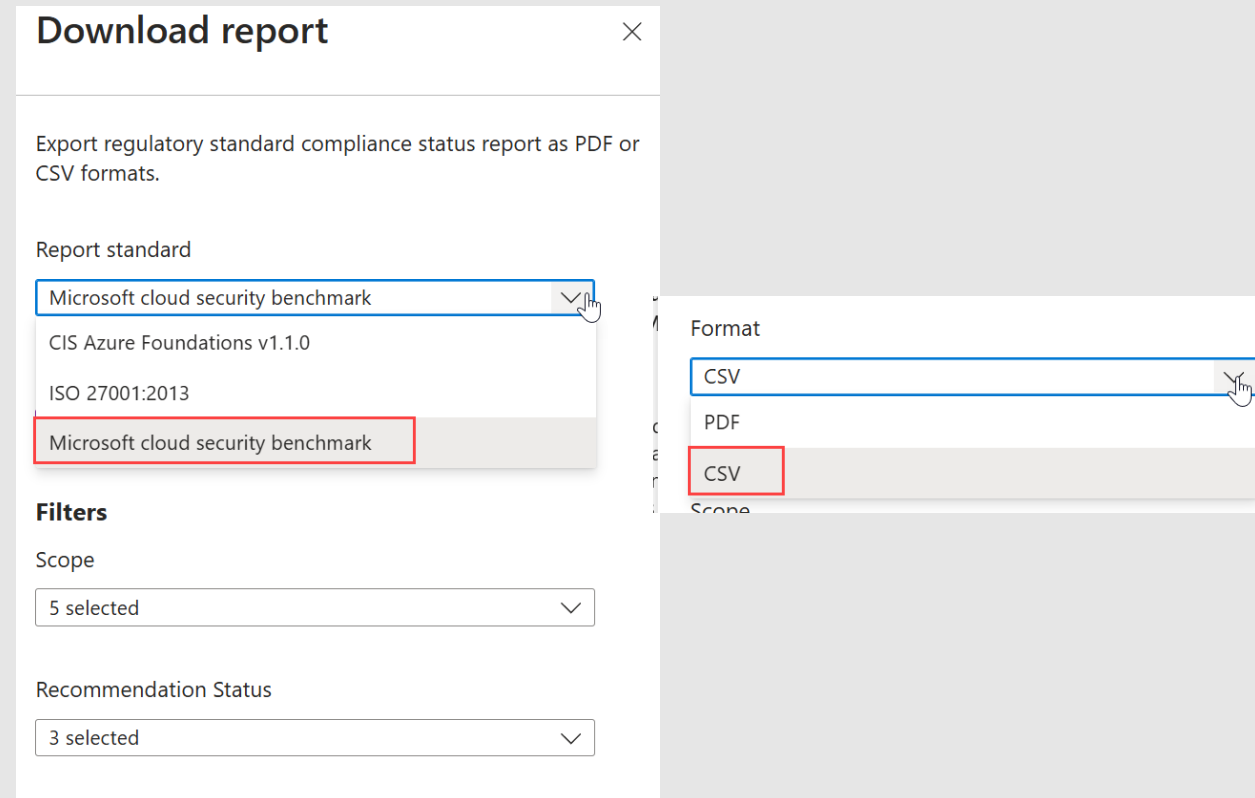
Instructions for Exporting Defender for Cloud
MCSB Regulatory Compliance Recommendations

Export Defender for Cloud: MCSB

In Defender for Cloud, select “**Regulatory compliance**” and then choose “**Download report**”:



Within “Download report,” ensure that the “**Microsoft Cloud Security Benchmark**” is selected and that the report format is set to **CSV**:



Script Export

(recommended)

Instructions for Exporting

- Defender for Cloud Recommendations
- Defender for Cloud MCSB Regulatory Compliance Recommendations

Script Artifacts

Filename	Purpose
ESA_MDC_DataExport.ps1	PowerShell script that downloads Defender for Cloud recommendations (MCSB regulatory compliance and MDC Classic Secure Score.)
MDC_Params.json	Parameter file containing export settings for MDC.
MCSB_Params.json	Parameter file containing export settings for MCSB.
MDC.kql	The KQL (Kusto Query Language) query executed by the script for MDC.
MCSB.kql	The KQL (Kusto Query Language) query executed by the script for MCSB.

<https://github.com/microsoft/esa>

Script Parameter File: MDC_Params.json

```
{
  "CSVFileName": "Export_MDC_Recommendations.csv",
  "QueryFile": "MDC.kql",
  "SubscriptionIds": ["*"],
  "_comments": {
    "CSVFileName": "Final CSV output file for Power BI import",
    "QueryFile": "KQL Query file. Do not change!",
    "SubscriptionIds": "Comma-separated subscription IDs or '*' for all available subscriptions. If you specify specific subscriptions, ensure you connect to the correct tenant (if you have access to multiple tenants) during script execution. If you don't have access to some subscriptions, the script will continue with the remaining ones."
  }
}
```

Parameter	Description
CSVFileName	The name of the file containing the exported MDC recommendations. This file needs to be imported into the Power BI report. You may change it, but it is recommended to keep the default value.
QueryFile	The file containing the KQL query. Do not modify.
SubscriptionIds	To export data for all available subscriptions, keep the default value '*'. Otherwise, specify the subscription IDs to be exported. Example: ["09b43e75...", 4fc2c46b...", ...] If you don't have access to some subscriptions, the script will continue with the remaining ones.

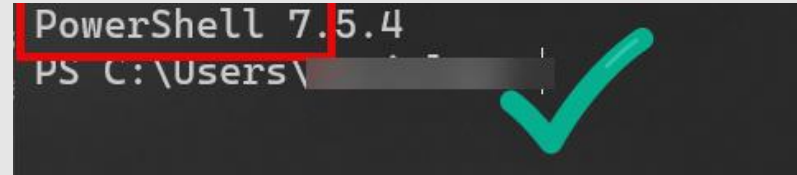
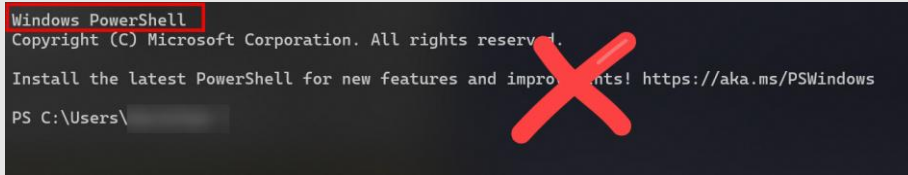
Script Parameter File: MCSB_Params.json

```
{
  "CSVFileName": "Export_MCSB_Compliance.csv",
  "QueryFile": "MCSB.kql",
  "SubscriptionIds": ["*"],
  "_comments": {
    "CSVFileName": "Final CSV output file for Power BI import",
    "QueryFile": "KQL Query file. Do not change!",
    "SubscriptionIds": "Comma-separated subscription IDs or '*' for all available subscriptions. If you specify specific subscriptions, ensure you connect to the correct tenant (if you have access to multiple tenants) during script execution. If you don't have access to some subscriptions, the script will continue with the remaining ones."
  }
}
```

Parameter	Description
CSVFileName	The name of the file containing the exported MCSB recommendations. This file needs to be imported into the Power BI report. You may change it, but it is recommended to keep the default value.
QueryFile	The file containing the KQL query. Do not modify.
SubscriptionIds	To export data for all available subscriptions, keep the default value '*'. Otherwise, specify the subscription IDs to be exported. Example: ["09b43e75...", 4fc2c46b...", ...] If you don't have access to some subscriptions, the script will continue with the remaining ones.

Script Execution - Requirements

- The script must be run in a **PowerShell 7.x** or Windows Terminal session.
Note: You may be using Windows PowerShell (5.x), which is not supported.
Installation instructions for PowerShell 7.x can be found here: [Installing PowerShell on Windows - PowerShell | Microsoft Learn](#)



- [Azure PowerShell](#) modules are required. The script will verify this during execution.
- Run the script with **standard user rights (non-admin console)**.
- The user running the script must have at least [Security Reader or Reader](#) role assigned to all subscriptions.
- PowerShell 7.x must run in [FullLanguage](#) mode (default) - not in constrained mode.
- The script **must not** be executed within an Azure Cloud Shell console.

Script Execution

```
.\ESA_MDC_DataExport.ps1 <<enter>>
```

If you run the script without any parameters, a help message will be displayed.

You need to run it with a specified parameter file, for example:

```
.\ESA_MDC_DataExport.ps1 .\MDC_Params.json <<enter>> (to download MDC classic recommendations)
```

```
.\ESA_MDC_DataExport.ps1 .\MCSB_Params.json <<enter>> (to download MCSB recommendations)
```

If there is an existing Azure session you may proceed with it or select 'N' to re-authenticate:

```
PS C:\ESA_MDC> .\ESA_MDC_DataExport.ps1 .\MDC_Params.json
You are currently authenticated as: u5[redacted]house.co | Tenant: ([redacted] House ([redacted]))
Do you want to continue with this session? (Y/N): |
```

Script Execution

You may see **yellow** warning messages after authentication. **These can be ignored!**

```
WARNING: To override which subscription Connect-AzAccount selects  
orLogin 00000000-0000-0000-0000-000000000000'. Go to https://go.r  
n.  
WARNING: Unable to acquire token for tenant '(redacted)'.  
inst tenant (redacted). User interactio
```

If you have access to multiple tenants, you will be prompted to select the desired tenant.

```
Available Tenants:  
[1] (redacted) 9c)  
[2] (redacted) ca0e)  
[3] (redacted)  
[4] (redacted) a0820b6d3)  
[5] (redacted) 77)  
  
Enter the number of the tenant you want to use: |
```

Script Execution

```
Current tenant: House ( )
Retrieving available subscriptions for the selected tenant ( )...
Querying subscription (1/5): 
Subscription 2 - Retrieved 0 records
Querying subscription (2/5): 
Subscription - Retrieved 59 records, remaining: 0
Querying subscription (3/5): 
Subscription - Retrieved 1000 records, remaining: 547
Subscription - Retrieved 547 records, remaining: 0
Querying subscription (4/5): 
Subscription - Retrieved 1000 records, remaining: 19
Subscription - Retrieved 19 records, remaining: 0
Querying subscription (5/5): 
Subscription - Retrieved 1000 records, remaining: 3284
Subscription - Retrieved 1000 records, remaining: 2284
Subscription - Retrieved 1000 records, remaining: 1284
Subscription - Retrieved 1000 records, remaining: 284
Subscription - Retrieved 284 records, remaining: 0

Overall Secure Score across 4 subscriptions: 65.25

Data export completed: Export_MDC_Recommendations_20250207_181605.csv (17.76 MB)
Total subscriptions queried: 5
Script execution time: 0h 1m 33s
```

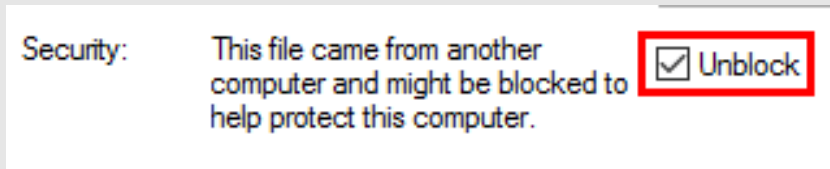
In the example above, 5 subscriptions were queried, and data was exported for 4 of them. The missing subscription (Yellow) was likely due to lack of access or other reasons. **This is not an error.** Depending on the number of subscriptions and the size of the data sets, the script may run for multiple hours. **The script calculates the overall secure score across the subscriptions at the end.**

Please share the exported CSV files and the Secure Score number with the Microsoft CSA.

(CSV files can be compressed (ZIP) to significantly reduce file size.)

Script Execution – Possible Errors

The script shared with you may be blocked on your machine because it is not yet signed. If this happens, you can check the .ps1 file properties and select "*Unblock*."



The script will notify you if any required PowerShell **Az** modules are missing and provide instructions on how to install them.

The script includes retry logic for some transient errors and will attempt to re-execute the calls after a delay. Status messages will be displayed if this occurs.

If any errors occur during the execution of the KQL query, they will be logged in a file with the **.failed** extension.

For any other issues, please contact your Microsoft CSA and provide screenshots and error files, if available.

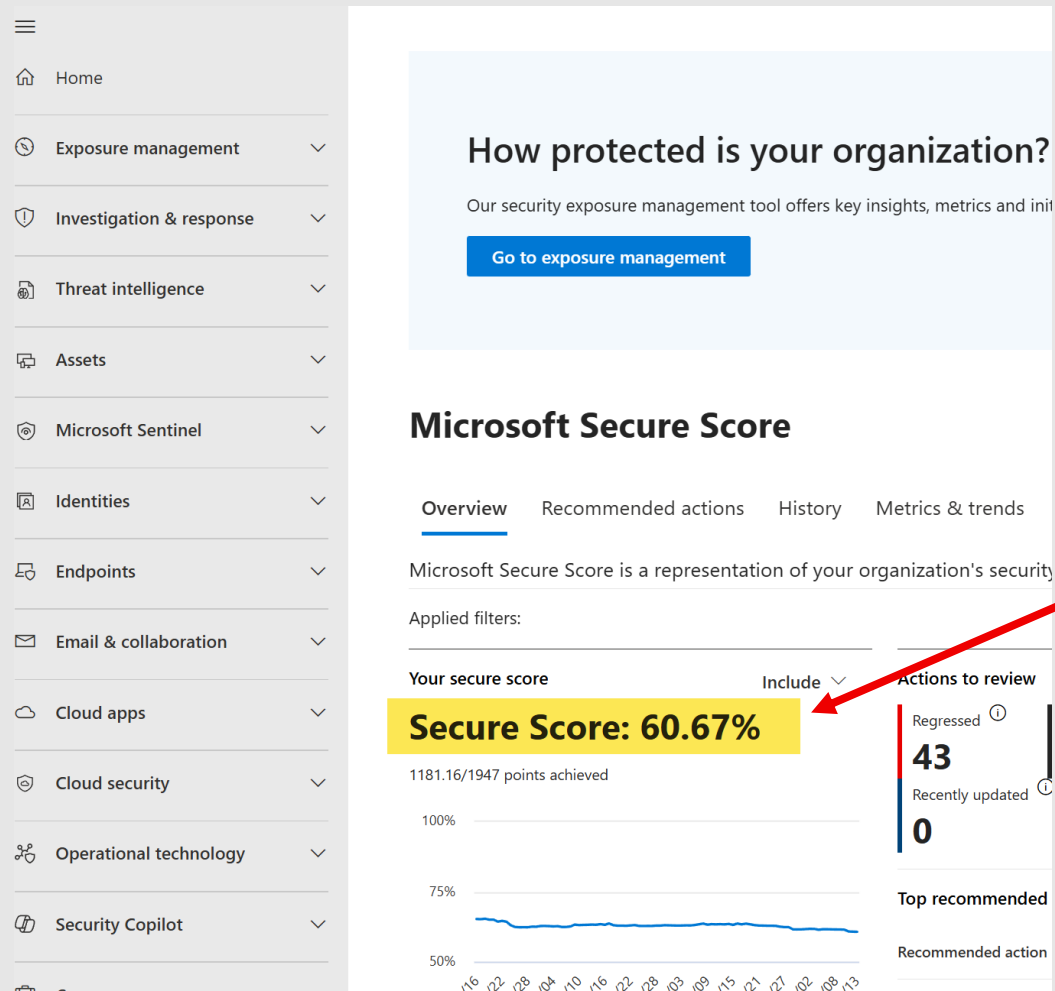
Export

Instructions for Exporting Microsoft Defender
XDR Recommendations (Microsoft Secure Score)

Microsoft Defender XDR recommendations need to be exported manually.

Export Microsoft Defender XDR

Navigate to: <https://security.microsoft.com/securescore>



Important:

Make a note of the Secure Score value, as this number will also be shared with Microsoft.

Export Microsoft Defender XDR

Select '**Recommended actions**' and click on "**Export**" to download the CSV report:



Zava Microsoft Defender

Microsoft Secure Score

Overview **Recommended actions** History Metrics & trends

Actions you can take to improve your Microsoft Secure Score. Score updates may take up to 24 hours.

[↓](#) Export

Rank	Recommended action
1	Turn on Microsoft Defender Antivirus real-time protection for Linux
2	Turn on Microsoft Defender Antivirus PUA protection in block mode for Linux
3	Identify Entra ID privileged accounts that are also privileged in Active Directory
4	Block rebooting machine in Safe Mode

Export

Instructions for Exporting Microsoft Purview
Improvement Actions

Microsoft Purview recommendations need to be exported manually.

Export Microsoft Purview

Navigate to: <https://purview.microsoft.com/compliancemanager/improvementactionspage>

The screenshot shows the Microsoft Purview Compliance Manager interface. The left sidebar contains navigation links: Home, Solutions, Agents, Learn, Usage center (preview), Settings, and Compliance Manager (highlighted). The main content area is titled 'Improvement actions' and includes a description: 'Improvement actions provide guidance on task completion which can improve your org's compliance score. Action points can take up to 24 hours to update.' Below this are buttons for 'Export actions', 'Update actions', 'Accept all updates', and 'Assign to user'. A filter section shows 'Regulations: Any', 'Solutions: Any', 'Groups: Any', 'Test Status: Any', 'Categories: Any', 'Testing type: Any', and 'Service: Any'. A table lists improvement actions with columns for checkboxes, action names, point counts, services, regulations, groups, and solutions.

☐	Improvement action	Point...	Service	Regulations	Grou...	Solutions
☐	Block outdated ActiveX controls	0/216	Microsoft 365	(8) Data protection b...	(2) Defaul...	Microsoft Def...
☐	Enable 'MIME Sniffing Safety Feature'	0/189	Microsoft 365	(7) Data protection b...	(2) Defaul...	Microsoft Def...
☐	Set default behavior for 'AutoRun' to 'E...	0/135	Microsoft 365	(5) Data protection b...	(2) Defaul...	Microsoft Def...
☐	Block process creations originating fro...	78.66/162	Microsoft 365	(6) Data protection b...	(2) Defaul...	Microsoft Def...

Export Microsoft Purview

Make sure that no filters are applied. All filter options must remain set to **Any**.

Select **Export actions** to download the 'ExportActions.xlsx' file.

Improvement actions

Improvement actions provide guidance on task completion which can improve your org's compliance score. Action points can take up to 24 hours to update. [Learn more about improvement actions](#)

↓ Export actions

↑ Update actions

✓ Accept all updates

📄 Assign to user

500 items

🔍 Search

☰ Group ▾

Filter

🔄 Reset

🔍 Filters

Regulations: Any ▾

Solutions: Any ▾

Groups: Any ▾

Test Status: Any ▾

Categories: Any ▾

Testing type: Any ▾

Service: Any ▾

Role type: Any ▾

Service Instances: Any ▾

Action update status: Any ▾

Assigned To: Any ▾

Final Checklist

Step	Status
All four files have been exported (three CSV, one Excel file) in accordance with the provided guidance. No files have been modified.	☒
If using manual export, ensure Defender for Cloud recommendations were exported from the classic view , not the default risk-based view.	☒
The Defender for Cloud Secure Score has been captured (e.g., via screenshot).	☒
The Microsoft Secure Score has been captured (e.g., via screenshot).	☒
All files have been compressed/zipped.	☒
All files, including screenshots with secure score values, have been uploaded to the secure file-sharing link provided by the CSA.	☒

Final Steps



- Please share the **four** downloaded files with Microsoft.
- Microsoft will provide you with a secure file transfer mechanism.
- Additionally, include the Defender for Cloud Secure Score and the Microsoft Secure Score values captured earlier.

Microsoft will complete the security assessment and present the results during the scheduled Teams meeting.



Thank you